Aurora Forecasting – Teacher Copy

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In this class, students learn about the key variables that influence aurora visibility, use simulated data to make a simple aurora forecast, and become familiar with real forecasts from the Bureau of Meteorology and NOAA.

Students require a computer/ tablet and access to the internet for this class.

Earth & space science curriculum link:

Analyse data and information to describe patterns, trends and relationships and identify anomalies (AC9S7I05 AC9S8I05).

Write and create texts to communicate ideas, findings and arguments for specific purposes and audiences, including selection of appropriate language and text features, using digital tools as appropriate (AC9S7I08 AC9S8I08).

Pre-requisite knowledge:

- Students should have a conceptual understanding of Earth's magnetic field.
- Students should have encountered the concept of the aurora before, but they do not need a prior understanding of what causes it.
- Students should be able to interpret trends in data and make logical conclusions from multiple data sources.

Aims:

- Understand what the aurora is and what causes it.
- Explain the three key variables that determine if the aurora will be visible.
- Consider the geographical conditions that allow the aurora to be viewed.
- Interpret scientific data to make and present an aurora forecast.
- Understand how to read real aurora forecasts from the BoM/ NOAA.

Activity 1: Understanding the aurora

Watch the following videos from the Bureau of Meteorology and answer the questions:

What is an aurora? : https://youtu.be/FpLd20_htF8?si=IZxycn8GHUl29wX

1. What is the name for the rings around the poles where auroras are concentrated?

Auroral ovals.

2. What are auroras called in the southern hemisphere?

Aurora australis.

3. What is the solar wind?

High energy particles (plasma, electrons and protons) released from the Sun.

4. How fast can the solar wind travel when the sun is active (during a solar storm)?

2000 km/s.

5. What protects Earth from the solar wind?

The magnetosphere.

6. What two gases in Earth's atmosphere interact with the solar wind?

Oxygen and nitrogen.

7. What determines the colour of the aurora?

The energy and location of the collisions between the solar wind and atoms of nitrogen and oxygen in the atmosphere.

Catching the aurora : https://youtu.be/0xUF82rVljE?si=PAnE9pQo5CDhmWBQ

8. What weather conditions are best for aurora viewing?

Dark night, little cloud cover, not a bright moon, no light pollution.

9. Give an example of a good aurora viewing location.

Dark beach or a hill.

10. Which compass direction should you face to view the southern lights?

South.

11. What times are best for aurora viewing?

10 pm – 2 am.

Activity 2: Making an aurora forecast

The visibility of the aurora is controlled by 3 geomagnetic variables. These are:

1. **Solar wind speed:** The speed of solar particles emitted from the sun.
2. **Kp index:** A measure of geomagnetic activity on a scale from 0 to 9.
3. **Bz:** The orientation of the solar wind's magnetic field.

(SWPC NOAA)

This data is collected by satellites positioned between Earth and the Sun. One of these is NASA's **DSCOVR** (Deep Space Climate Observatory). This satellite is positioned approximately 1.5 million kilometres from Earth and monitors the solar wind in real-time (NOAA NESDIS). The data collected is used to make aurora forecasts and alerts.

You can view the Real Time Solar Wind data here: https://www.swpc.noaa.gov/products/real-time-solar-wind

By default, the data for the past 7 days is shown however there are options underneath the graphs to change the time period and type of data displayed.

Figure 1 shows where the Bz, wind speed and Kp data is located.

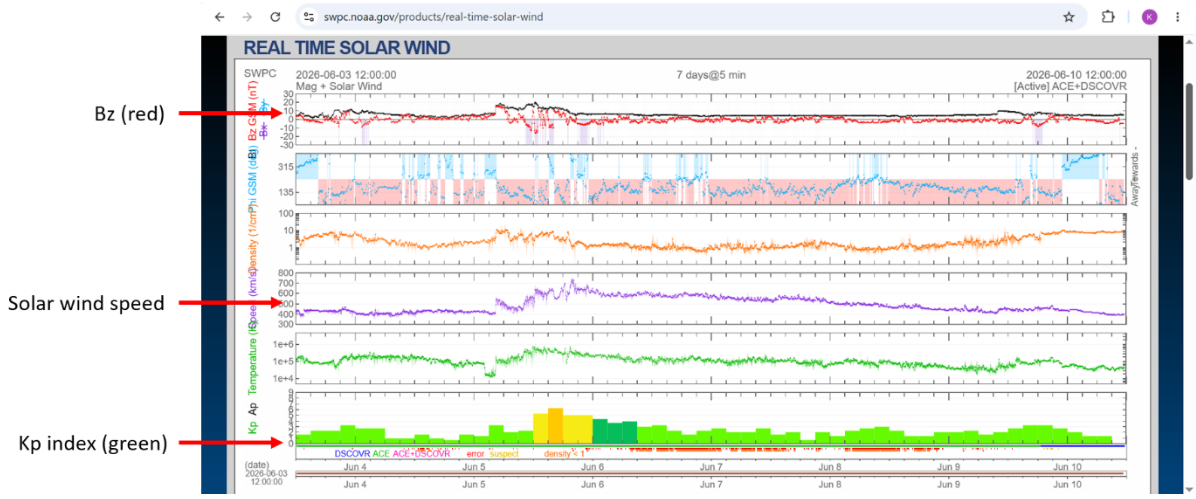

Figure 1: Labelled real time solar wind data for the period of 3 June – 10 June 2026 (SWPC NOAA, https://www.swpc.noaa.gov/products/real-time-solar-wind).

The graphs for Bz, solar wind speed and Kp index are labelled.

Encourage students to explore the real time solar wind webpage and change the data types and time period displayed.

For the aurora to be visible, all the following conditions must be met:

1. Solar wind speed must be fast
2. Kp must be high
3. Bz must be negative (southward)

(69Nord, 2025)

Your teacher will now arrange you into small groups.

Imagine you are a team of scientists working at the Australian Space Weather Forecasting Centre.

You have been provided with data about the geomagnetic conditions for 3 different times (**Table 1**) and descriptions of 3 different locations in the Hobart area (**Table 2**). You can also look these locations up on Google Maps.

Your task is to study the data in the tables and decide:

a. Will the aurora be visible at each time?
b. When would be the best time to view the aurora?
c. Which location would be best to view the aurora from?

You should compile your ideas as a short weather forecast-like report.

Fully justify your choice of aurora-viewing time and location.

You will then present your group's forecast to the rest of the class.

Time	Kp	Solar wind (km/s)	Bz (nT)
6 pm	3	350	+2
9 pm	5	500	-4
12 am	7	650	-8

Table 1: The solar wind, Kp index and Bz value for three different times.

Location	Description of Location
Short Beach	The closest beach to Hobart city (15 minutes' walk away). Within the Sandy Bay suburb. Faces southeast.
Taroona Beach	A beach 15 minutes' drive out of Hobart city with an unobstructed view to the South.
Franklin Square	A park in the heart of Hobart city.

Table 2: Descriptions of 3 different locations around Hobart.

Arrange the class into groups of 3 or 4 students. Give them 20 minutes maximum to prepare their forecasts.

Students should predict that **12 am** is the best time to view the aurora because of the following reasons:

- Latest time = darkest skies
- Highest Kp value
- Fastest solar wind speed
- Strongly negative Bz value

Students should choose **Taroona Beach** as the best aurora viewing location because:

- It is the furthest from the city so has the least amount of light pollution
- The beach has a clear view to the south which is where auroras are seen

Below is an example aurora forecast:

"We predict aurora visibility at 12 am because the Kp is rising to 7, solar wind speed is high and Bz is strongly negative. Taroona beach will be the best location to view the aurora as it has the darkest skies and a clear view to the south."

Extension question: If you finish your aurora forecast before it is time to present to the class, research where your closest aurora viewing location is.

Good aurora viewing locations around Hobart are:

- Mt Nelson Signal Station
- Rosny Hill Lookout
- Howrah Beach
- Seven Mile Beach
- Taroona Beach
- Park Beach
- Mt Wellington (kunanyi) summit
- Howden boat ramp

Spots around South Arm include:

- Cremorne Beach
- Clifton Beach
- Goat Bluff Lookout
- Calvert's Beach
- Gypsy Bay / Primrose Sands

Reference: Strikis, A. (2026). *Complete Guide to the Southern Lights in Tasmania | Aurora Australis*. Available at: https://lapoftasmania.com.au/southern-lights-in-tasmania/

Activity 3: Reading BoM and NOAA aurora forecasts

We will now look at real aurora forecasts and learn how to read these.

Navigate to the Australian Space Weather Forecasting Centre:
https://www.sws.bom.gov.au/Aurora

This webpage displays the current aurora conditions, as well as current values for the Kp index, solar wind speed and Bz value. These are labelled on **Figure 2**.

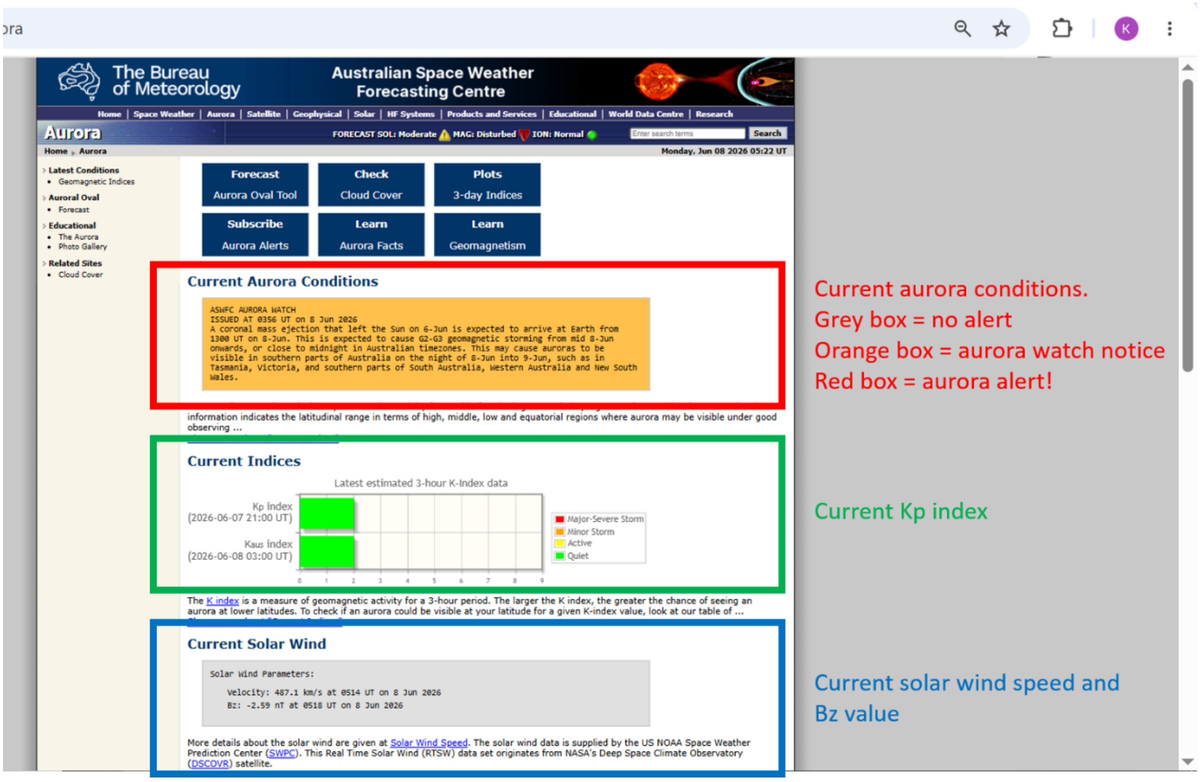

Figure 2: Labelled aurora conditions webpage from the BoM's Australian Space Weather Forecasting Centre for 8 June 2026 (SWS BoM, https://www.sws.bom.gov.au/Aurora).

Current aurora conditions (red box), current Kp index (green box), current solar wind and Bz value (blue box).

<u>Current aurora conditions:</u>

The colour of the aurora conditions text box changes according to the aurora alert level:

- **Grey box:** no current aurora alert.
- **Orange box:** aurora watch notice. Issued when solar wind conditions are favourable for auroras occurring in the next 1-3 days (BoM, 2018).
- **Red box:** aurora alert. Issued when there is a high chance that an aurora will be visible <u>now</u> (BoM, 2018).

Look at the current conditions and answer the following questions:

1. Is there currently an aurora alert in place?

2. What is the current Kp index? Are conditions quiet, active or stormy?

3. What are the current values of solar wind speed (velocity) and Bz value?

Students should record the current auroral and solar wind conditions on the BoM SWS page.

Short-term aurora forecasting data is also available from the NOAA.

Navigate to the NOAA's 30-minute aurora forecasting webpage: https://www.swpc.noaa.gov/products/aurora-30-minute-forecast

The webpage contains animations that show the extent of the auroral oval at both poles for the next 24 hours based on current forecasts (see **Figure 3**). The oval is coloured according to aurora probability.

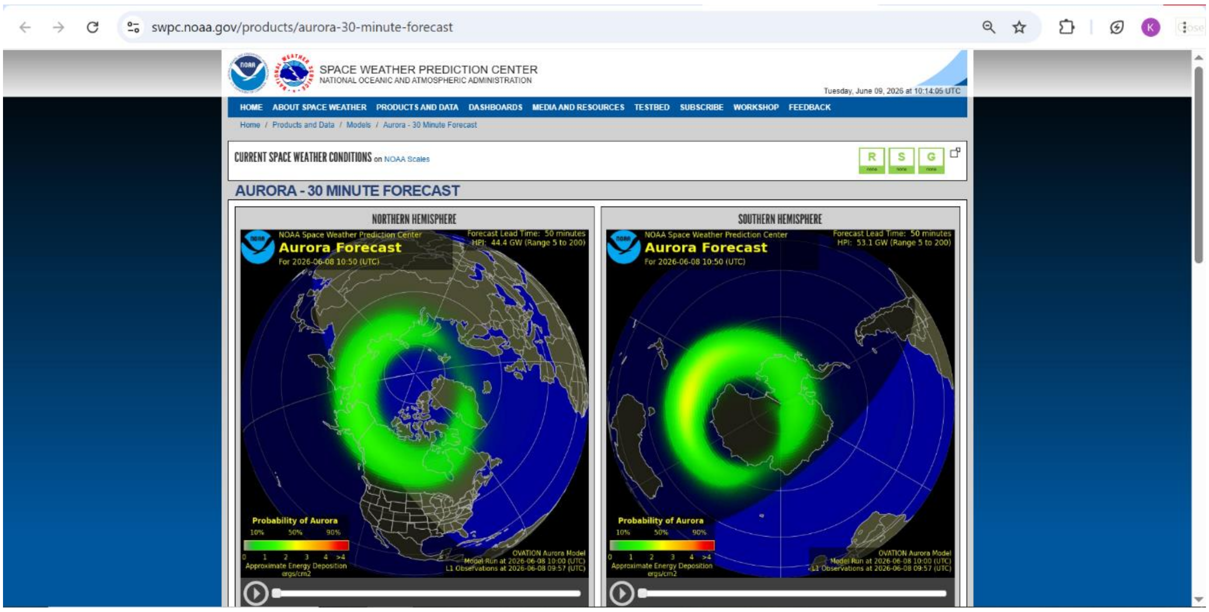

Figure 3: NOAA's 30-minute Aurora Forecast webpage for 9 June 2026 (SWPC NOAA, https://www.swpc.noaa.gov/products/aurora-30-minute-forecast).

Auroral oval animation for the northern hemisphere (left) and southern hemisphere (right).

Press the play button in the bottom left corner of each animation to watch how the auroral oval changes over the course of the next day.

1. Will the auroral oval in the southern hemisphere reach Tasmania within the next 24 hours?

2. Are there any locations with a high (> 90%) chance of an aurora within the next 24 hours?

Students should watch the current auroral oval animations to see if the auroral oval extends to Tasmania and if there are any red oval areas (> 90% auroral visibility) within the next 24 hours. If there are any red areas, students should note which countries/ regions these are.

Now that you know how to read an aurora forecast and which locations are best for aurora viewing, you are now fully equipped to go out aurora hunting! Good luck on your search!

References

Bureau of Meteorology (2017). *Ask the Bureau: What is an aurora?* Available at: https://youtu.be/FpLd20_htF8?si=dw4fhw4WoMX7ec9e

Bureau of Meteorology (2018). *BOM Weather Guide: catching the aurora.* Available at: https://youtu.be/0xUF82rVljE?si=ocMK5xvOb0BIfWp8

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69Nord (2025). *Understanding the indicators of Northern Lights viewing apps.* Available at: https://69nord.com/en/announcement/understanding-the-indicators-of-northern-lights-viewing-apps/

Space Weather Prediction Center National Oceanic and Atmospheric Administration (SWPC NOAA). *Aurora - 30 Minute Forecast.* Available at: https://www.swpc.noaa.gov/products/aurora-30-minute-forecast

Space Weather Prediction Center National Oceanic and Atmospheric Administration (SWPC NOAA). *Real Time Solar Wind*. Available at: https://www.swpc.noaa.gov/products/real-time-solar-wind

Space Weather Prediction Center National Oceanic and Atmospheric Administration (SWPC NOAA). *Space Weather Glossary*. Available at: https://www.swpc.noaa.gov/content/space-weather-glossary

This resource was developed with assistance from ChatGPT (OpenAI, 2026), used to generate and refine instructional ideas.